AD44/AI44

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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

BACKLOG SUBJECT EXAMINATIONS – FEBRUARY 2024

Program	:	**B.E. - Artificial Intelligence and Data Science / Artificial Intelligence and Machine Learning**	Semester	:	**IV**
Course Name	:	**Theory of Computation**	Max. Marks	:	**100**
Course Code	:	**AD44 / AI44**	Duration	:	**3 Hrs**

Instructions to the Candidates:

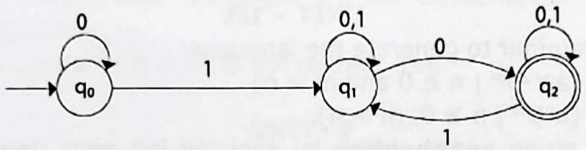
- Answer one full question from each unit.

UNIT – I

1. a) Design a DFA that accepts strings from L= {ω | ω is in {a,b}* and does not contain either aa or bb }. Show that the string abab is in L. CO1 (06)

   b) Give the difference between DFA, NFA and ε-NFA. CO1 (06)

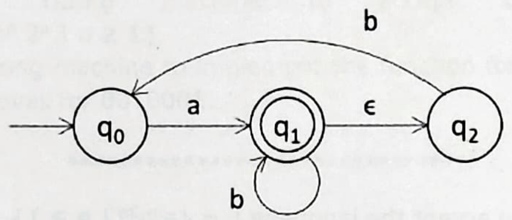
   c) Convert the following NFA to DFA: CO1 (08)



2. a) Draw the DFA that accepts language: CO1 (08)
   $L = \{w \in \{a, b\}^* : N_a(w) \bmod 3 > N_b(w) \bmod 3\}$

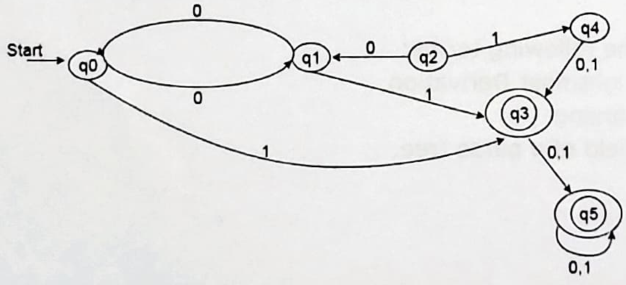
   b) Prove that A language L is accepted by some ε-NFA if and only if L is accepted by some DFA. CO1 (06)

   c) Convert the following ε-NFA into its equivalent DFA. CO1 (06)



UNIT – II

3. a) Minimize the following DFA using table filling method. CO2 (08)



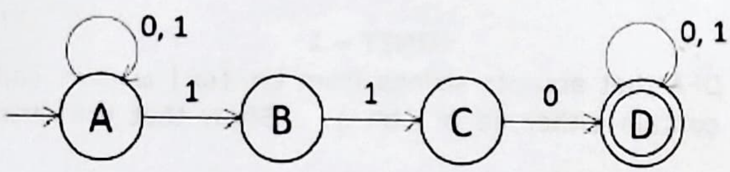
b) Prove that regular languages are closed under union and concatenation. CO2 (06)

c) State and prove if L=L(A) for some DFA A, then there is a regular expression R in such that L = L(R). CO2 (06)

4. a) Define Regular expression. And solve the following:- CO2 (06)
   i. Obtain a regular expression for Strings of a's and b's whose last and second last symbols are same.
   ii. Obtain a regular expression for $L = \{a^n b^m : n + m \text{ is even}\}$

b) Show that $L = \{a^n b^n \mid n \geq 0\}$ is not regular. CO2 (06)

c) Convert the following automata to regular expression by eliminating states. CO2 (08)



UNIT – III

5. a) Obtain a grammar to generate the language: CO3 (08)
   i) $L = \{a^{n+2} b^m \mid n \geq 0 \text{ and } m > n\}$
   ii) $L = \{a^n b^m \mid n \geq 0, m > n\}$

b) Obtain the string **aaabbabbba** by applying left most derivation and the parse tree for the grammar given below. Is it possible to obtain the same string again by applying leftmost derivation but by selecting different productions? CO3 (08)

S->aB|bA

A->aS|bAA|a

B->bS|aBB|b

c) Eliminate all ε production for the grammar: CO3 (04)

$S \rightarrow ABC \mid bD$

$A \rightarrow BC \mid b$

$B \rightarrow b \mid \varepsilon$

$C \rightarrow c \mid \varepsilon$

$D \rightarrow d$

6. a) Obtain a PDA to accept the language $L = \{a^n b^{2n} \mid n \geq 1\}$. CO3 (08)

b) Prove that the following grammar is ambiguous: CO3 (06)

S->AB|aaB

A->a|Aa

B->b

c) Define the following terms: CO3 (06)
   i) Rightmost Derivation
   ii) Parsing
   iii) Yield of a parse tree.

UNIT – IV

7.	a)	If L1 and L2 are context free languages then prove that family of context-free-languages are closed under union and concatenation operations.	CO4	(06)
	b)	Obtain the grammar in CNF form: S->a|aA|B|C, A->aB|ϵ B->aA|ϵ C->cCD D->ddd	CO4	(08)
	c)	State and prove the pumping lemma for context-free language.	CO4	(06)
8.	a)	Obtain the grammar in CNF form: S->aA|a|B|C A->aB|ϵ B->aA C->cC D->abd	CO4	(06)
	b)	Prove that context free languages are not closed under intersection and complementation.	CO4	(08)
	c)	Prove that L = $\{a^{n!} \mid n \geq 0\}$ is not context free.	CO4	(06)

UNIT- V

9.	a)	Explain different types of Turing Machines.	CO5	(06)
	b)	Define the following: i) Recursively enumerable language ii) Decidable Language.	CO5	(04)
	c)	Define Turing Machine. Obtain a Turing Machine to accept the language L=$\{0^n 1^n \mid n>=1\}$.	CO5	(10)
10.	a)	Obtain a Turing machine to accept the language L(M) = $\{0^n 1^n 2^n \mid n \geq 1\}$.	CO5	(10)
	b)	Design a Turing machine to implement the function for multiplication. Show the moves for 0010001.	CO5	(10)

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